the closest entry to a target tuple, like (100$, 4.5 rating). Even more we will be able to ask for the `N` closest entries to that target.

In this and the following chapter we are going to provide three data structures, three containers, that allow for efficient nearest neighbors queries. But not just that: they will provide a few special operations:

- Retrieving the `N` closest points to a target point (not necessarily in the container).
- Retrieving all the points in the container within a certain distance from a target point (geometrically interpretable as all points within a hyper-sphere).
- Retrieving all the points in the container within a range (all points lying inside a hyper-rectangle, or a semi-hyperspace).

Let's now briefly introduce the three structures we will describe in these chapters.

- **Kd-tree**: a kd-tree is a special binary tree in which every non-leaf node represents a splitting hyper-plane that divides the k-dimensional space into two half-spaces. Points to the left of this splitting hyper-plane are stored in the left subtree of the node and points to the right of the hyper-plane are stored in the right subtree.

  We'll focus on kd-trees in chapter 9.

- **R-tree**: The *R* here is for rectangle: an R-tree groups nearby points and defines the minimum bounding box (i.e. hyper-rectangle) containing them. Points in the